

Offline Reinforcement Learning-based Adaptive Impedance Control for Variable Force Tracking in Robotic Ultrasound Scanning

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Abstract—Robotic ultrasound scanning (RUSS) has become a promising solution to standardize ultrasound (US) examinations and relieve the workload of sonographers. In many practical RUSS applications, the probe must track a time-varying force profile, especially when scanning patients with musculoskeletal pain, tumors, or other localized lesions. In such cases, conventional adaptive impedance control (AIC) is difficult to tune and often fails to generalize across different patients and anatomical regions. Moreover, directly learning a force control policy online by trial-and-error on patients is clinically unacceptable due to safety concerns. To address these challenges, we introduce an offline reinforcement learning based adaptive impedance control scheme (ORL-AIC) for variable force tracking in RUSS. The proposed controller learns a near-optimal policy from a limited set of pre-collected interaction samples, without requiring additional online exploration on patients. Phantom and in vivo experiments on a real RUSS platform demonstrate that ORL-AIC achieves reduced overshoot and improved tracking accuracy compared with conventional AIC baselines, enabling safer and more patient-specific robotic US scanning.

Index Terms—robotic ultrasound scanning, offline reinforcement learning, variable force tracking, adaptive impedance control

I. INTRODUCTION

Robotic ultrasound scanning (RUSS) has recently attracted increasing attention as a means to standardize ultrasound (US) examinations and to reduce the dependence on highly experienced sonographers [1]. In a typical RUSS workflow, a pre-specified scan trajectory is first generated by considering the anatomical structures of interest, which can be reconstructed from RGB-D sensing [2], [3] or CT/MRI data [4]. The robot then drives the US probe to follow this path and applies a suitable contact force normal to the tissue

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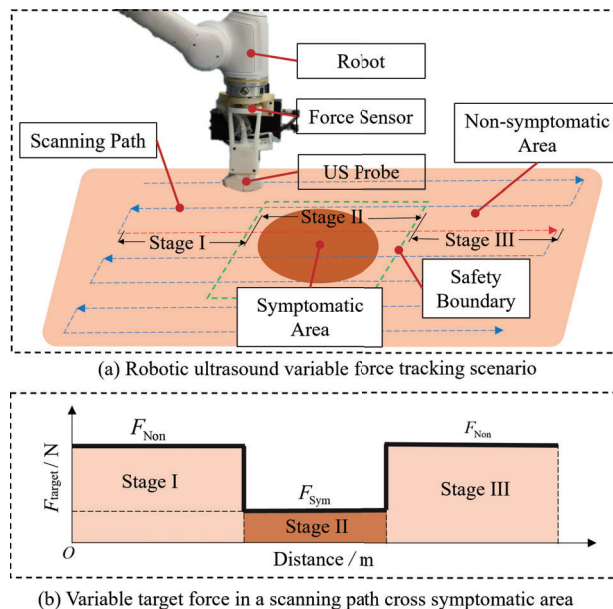


Fig. 1. Robotic variable force tracking in medical US scanning.

surface, as sketched in Fig. 1(a). However, the force that a patient can tolerate is not uniform along the scan path. It may change substantially between healthy and symptomatic regions, as illustrated in Fig. 1(b). Lesion areas (e.g., with musculoskeletal pain or tumors) are typically more sensitive, so the admissible probe pressure must be reduced when the probe passes over these regions. Consequently, the desired interaction force becomes a position-dependent signal, and the controller has to track a varying force reference when the probe moves from non-symptomatic to symptomatic tissue. Excessive overshoot during this variable force tracking can cause discomfort or even iatrogenic injury, whereas poor tracking accuracy degrades image quality. Hence, there is a strong clinical motivation to develop a force controller that can rapidly follow these time-varying targets while maintaining small overshoot and high tracking precision.

Force control in robotic interaction is most commonly formulated within the impedance control (IC) framework [5]. Despite its theoretical elegance, conventional IC often exhibits limited robustness in practical applications. This limitation

primarily stems from the inevitable uncertainties present in the robot dynamic model, which typically preclude the precise realization of the target impedance. To mitigate these challenges, a variety of IC-based methods have been proposed [6]–[9]. Among these, adaptive impedance control (AIC) [6]–[8] has emerged as an effective solution, offering significant improvements over traditional IC approaches. However, the hyperparameters of AIC are challenging to set for unstructured contact objects like human skin, and improperly configured AIC algorithms can lead to significant overshoot and poor tracking accuracy, which will cause patient discomfort and, in severe cases, medical accidents.

With the rapid progress of reinforcement learning (RL), many recent works have investigated its use for tackling adaptive control tasks [10]–[13]. However, in medical scenarios, applying RL in an online manner is problematic because the robot has to interact directly with patients while collecting data and updating its policy, which can introduce considerable safety risks [10], [14]. To circumvent this issue, offline RL (ORL) has been proposed as an attractive alternative: it learns policies entirely from previously gathered datasets, without further real-time interaction with the environment [11], [15]. Such a training paradigm is especially suitable for domains where exploration is expensive or hazardous, as in medical robotics.

In this work, we develop an offline reinforcement learning based adaptive impedance control scheme (ORL-AIC) for variable force tracking in RUSS. The proposed controller learns a near-optimal policy from a finite collection of pre-recorded robotic scanning force data. We first revisit the structure of AIC and analyze its behavior under the condition of zero steady-state force error. On this basis, the AIC regulation process is reformulated as a Markov decision process (MDP). Using the RUSS platform, we then acquire 40 sets of force tracking trajectories generated by non-optimal controllers and employ ORL to iteratively improve the policy. The resulting variable-force tracking policy is evaluated on both phantom and in vivo experiments, which confirm the effectiveness of the proposed approach. To the best of our knowledge, this is the first ORL-driven adaptive variable impedance force tracking method that derives a near-optimal policy directly from real robotic scanning data, indicating the broader potential of ORL in medical robotic applications.

II. TECHNICAL BACKGROUND

A. Impedance Control and Adaptive Impedance Control

The impedance control law establishes a desired force-motion relationship through:

$$M\ddot{x} + D\dot{x} + K(x - x_{\text{ref}}) = F_{\text{ext}} \quad (1)$$

where M , D , and K define the inertia, damping, and stiffness of the virtual impedance, x is the end-effector position, x_{ref} is the reference trajectory, and F_{ext} is the external force.

Adaptive impedance control refines traditional methods by adjusting parameters to handle uncertainties, as shown in:

$$M\ddot{x} + \hat{D}\dot{x} + \hat{K}(x - x_{\text{ref}}) = F_{\text{ext}}, \quad (2)$$

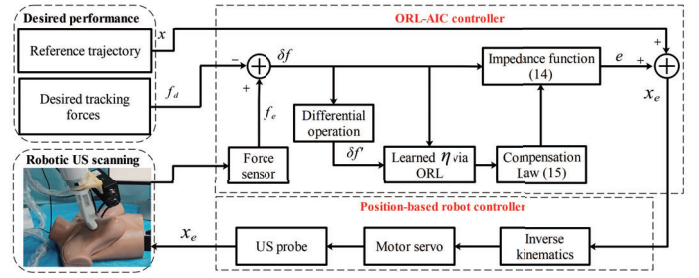


Fig. 2. ORL-AIC control framework.

where $\hat{D}$ and $\hat{K}$ are adaptively updated estimates of damping and stiffness.

B. Markov Decision Processes and Reinforcement Learning

RL problems are commonly formalized as MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, d_0, R, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ denote the state and action spaces, $T(s_{t+1} | s_t, a_t)$ is the transition kernel, d_0 is the initial-state distribution, R is the reward function, and $\gamma \in (0, 1]$ is the discount factor.

The objective is to find a policy π that maximizes the expected discounted return

$$J_\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \quad (3)$$

For a given policy π , the state-action value function (critic) is defined as

$$Q^\pi(s, a) = \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right], \quad (4)$$

which satisfies the Bellman equation [16]

$$Q^\pi(s_t, a_t) = \mathbb{E} [r(s_t, a_t) + \gamma Q^\pi(s_{t+1}, a_{t+1})]. \quad (5)$$

Consequently, the optimal critic function is defined as,

$$Q^*(s, a) = \max_{\pi} Q^\pi(s, a). \quad (6)$$

The optimal policy π^* can be obtained via a greedy strategy with the optimal critic function $Q^*(s, a)$, as follows,

$$\pi^*(s) = \arg \max_{a \in \mathcal{A}} Q^*(s, a) \quad (7)$$

III. PROPOSED ORL-AIC FOR ROBOTIC VARIABLE FORCE TRACKING

In clinical US scanning, variable force tracking is specifically designed to address the unique needs of certain patients. For instance, when patients experience musculoskeletal pain or tumors, a symptomatic area may exist where the tissue is more sensitive. The contact force that the patient can tolerate in this area is significantly lower; otherwise, it may lead to pain or even injury. To ensure patient safety and obtain high-quality US images, it is necessary to dynamically adjust the target tracking force based on the patient's actual condition. The variable force tracking function of the RUSS system can precisely meet the individual needs of patients, enhance

its generalization performance, and thus demonstrate superior performance in practical applications, providing patients with a higher-quality medical survive.

For variable force tracking, we only consider the component of the interaction force that is normal to the tissue surface, i.e., along the contact direction between the probe and the tissue. The impedance behavior along this axis can be described by

$$m_d \ddot{\Delta x_r} + b_d \dot{\Delta x_r} + k_d \Delta x_r - f_m + f_d = 0, \quad (8)$$

where m_d , b_d , and k_d denote the desired inertia, damping, and stiffness parameters, respectively. The term f_m is the force measured during contact, and f_d is the desired (reference) contact force. The quantity $\Delta x_r = x_e - x$ represents the position tracking error, with x_e the current probe position and x the commanded reference position.

The patient's skin is modeled as a position-dependent spring to describe its dynamic characteristics:

$$k_{\text{env}}(x_e - x_{\text{env}}) = f_{\text{env}}, \quad (9)$$

where k_{env} characterizes the varying stiffness of the tissue and x_{env} denotes the corresponding undeformed (rest) position.

Within the workspace, the trajectory tracking error Δx_r can be represented in the frequency domain as:

$$m_d \Delta x_r(z) z^2 + b_d \Delta x_r(z) z + k_d \Delta x_r(z) = \Delta f(z), \quad (10)$$

where z denotes the complex frequency variable.

The connection between the force tracking error Δf and the position deviation Δx_r can be modeled as a second-order dynamics with transfer function $G(z) = [m_d z^2 + b_d z + k_d]^{-1}$. Based on this relation, the controller adjusts the impedance parameters to compute a corrective displacement Δx_r , which is then superimposed on the reference trajectory. Hence, the commanded position becomes $x = x_e + \Delta x_r = x_e + \Delta f(z) G(z)$. Under position-servo control, we have $x = x_e$. The interaction force f_{env} is obtained from the force sensor, so that $f_e = f_{\text{env}}$. Therefore, the force tracking error can be rewritten as

$$\begin{aligned} \Delta f(z) &= f_e - f_d = k_{\text{env}}(x - x_{\text{env}}) - f_d \\ &= k_{\text{env}}(x_e + G(z)\Delta f(z) - x_{\text{env}}) - f_d. \end{aligned} \quad (11)$$

By substituting the transfer function $G(z) = [m_d z^2 + b_d z + k_d]^{-1}$ into equation (11), the force tracking error can be reformulated as:

$$\Delta f(z) = \frac{(m_d z^2 + b_d z + k_d)[k_{\text{env}}(x_e - x_{\text{env}}) - f_d]}{m_d z^2 + b_d z + k_d - k_{\text{env}}}. \quad (12)$$

Based on equation (12), the steady-state force tracking error is obtained by taking the limit as $z \rightarrow 0$:

$$\Delta f_{ss} = \lim_{z \rightarrow 0} \Delta f(z) = \frac{k_d \cdot [k_{\text{env}}(x_e - x_{\text{env}}) - f_d]}{k_d - k_{\text{env}}}. \quad (13)$$

To ensure zero steady-state force tracking error, the following condition must be satisfied: either $k_{\text{env}}(x_e - x_{\text{env}}) = f_d$ or $k_d = 0$. However, in the context of this work, both the environment's stiffness k_{env} and its rest position x_{env} are unknown a priori. As a result, any variation in the environment's properties leads to a corresponding change in the steady-state

error. Notably, setting the stiffness gain k_d to zero yields $\Delta f_{ss} = 0$ as a steady-state equilibrium.

Along the Z -axis, the AIC scheme can be divided into two operating modes. In the first mode (free-space control), the probe has not yet contacted the tissue. Once contact with the patient's skin is established, the controller switches to the second mode (contact-space control), in which variable force tracking is carried out. The corresponding adaptive force control law [6], [7] is given by

$$\begin{cases} m_d \ddot{\Delta x_r} + b_d \dot{\Delta x_r} + k_d \Delta x_r = -f_d & (\text{free space}) \\ m_d \ddot{\Delta x_r} + b_d (\dot{\Delta x_r} + \Xi) = f_e - f_d & (\text{contact space}) \end{cases}, \quad (14)$$

where Ξ is a time-varying compensation term updated online from the force tracking error according to

$$\Xi(t) = \Xi(t - \lambda) + \eta \cdot \frac{f_d(t - \lambda) - f_e(t - \lambda)}{b_d}, \quad \eta > 0, \quad (15)$$

with η denoting the adaptation gain and λ the sampling period of the controller. Taking the compensation term in (15) into account, the damping ratio ζ of the AIC system can be expressed as [8]

$$\zeta = \frac{b_d}{2\sqrt{m_d(1 + \eta)k_{\text{env}}}}. \quad (16)$$

Eq. (16) shows that increasing the update gain η reduces the damping ratio ζ , which tends to make the closed-loop response more oscillatory during contact. In contrast, choosing a smaller η results in higher damping and a more sluggish but stable behavior. Hence, using a single fixed value of η over the entire scanning procedure is generally undesirable.

To address this issue and ensure both accurate force tracking and safe interaction (i.e., avoiding overshoots), we propose a learning-based control approach termed ORL-AIC. This method adaptively regulates η using our previously developed BOREL method [11]. In this framework, the Broad Learning System (BLS) [17] is utilized to approximate the state-action value function (critic) in the ORL algorithm. The approximated critic function is expressed as:

$$\begin{aligned} \hat{Q}^\pi(s_t, a_t) &= \sum_{k=1}^n \phi_k(s_t, a_t)^T \beta_k + \sum_{l=1}^m \varphi_l(s_t, a_t)^T \beta^l \\ &= \psi(s_t, a_t) \beta_n^m, \end{aligned} \quad (17)$$

where the unified basis function $\psi_k(s_t, a_t)$ is defined as:

$$\psi_k(s_t, a_t) = \begin{cases} \phi_k(s_t, a_t), & \text{if } k \leq n \\ \varphi_{k-n}(s_t, a_t), & \text{if } k > n \end{cases}$$

and the full feature vector is:

$$\psi(s_t, a_t) = [\psi_1(s_t, a_t), \psi_2(s_t, a_t), \dots, \psi_{n+m}(s_t, a_t)]^T.$$

The weight vector is denoted as $\beta_n^m = [\beta_1, \dots, \beta_n, \beta^1, \dots, \beta^m]$. Here, $\phi_k(\cdot)$ represents the broad feature nodes, while $\varphi_l(\cdot)$ denotes the enhancement nodes, as described in the BLS architecture [17]. The integers n and m

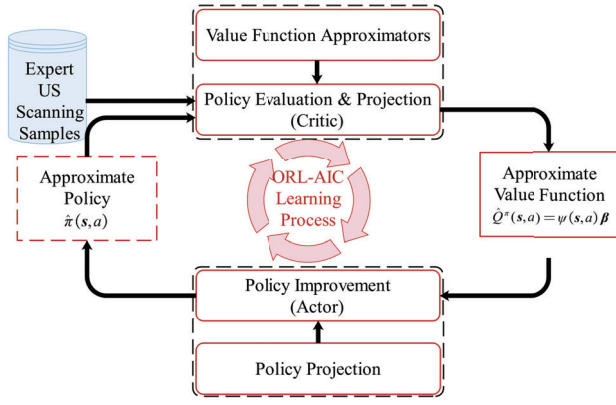


Fig. 3. ORL-AIC learning process

correspond to the number of feature and enhancement node groups, respectively.

The parameters β_n^m can be efficiently estimated using the Least-Squares Fixed-Point Approximation method [11], yielding the following solution:

$$\beta_n^m = \left(\Psi^T (\Psi - \gamma \Psi') \right)^{-1} \Psi^T R, \quad (18)$$

where

$$\begin{aligned} \Psi &= [\psi(s_1, a_1), \psi(s_2, a_2), \dots, \psi(s_N, a_N)]^T, \\ \Psi' &= [\psi(s'_1, \pi(s'_1)), \psi(s'_2, \pi(s'_2)), \dots, \psi(s'_N, \pi(s'_N))]^T, \\ R &= [r_1, r_2, \dots, r_N]^T, \end{aligned} \quad (19)$$

and π denotes the policy obtained at the present iteration. Policy improvement is then carried out by applying a greedy update with respect to the approximated critic:

$$\pi(s) = \arg \max_{a \in \mathcal{A}} \psi(s, a) \beta_n^m. \quad (20)$$

This policy improvement process is repeated iteratively. After each iteration, the target matrix Ψ' is updated based on the new policy. The learning process continues until convergence is achieved, i.e., when the change in the weight vector falls below a predefined threshold: $\|\beta_n^m - (\beta_n^m)'\|_2 < \varepsilon$, where ε is a manually set stopping criterion. The learning workflow of the proposed ORL-AIC approach is illustrated in Fig. 3.

The MDP state is constructed from two variables, namely $s = [\delta f, \delta f']$, where $\delta f = f_e - f_d$ denotes the instantaneous force tracking error, and $\delta f'$ is its temporal derivative (i.e., the rate of change of δf). The action space contains four discrete choices, each specifying a particular adaptation gain η ; thus we define $\mathcal{A} = \{0.001, 0.005, 0.01, 0.05\}$. The reward function is designed to discourage large overshoot and slow convergence, while encouraging stable and accurate force regulation. In our setting, force fluctuations within ± 0.5 N are regarded as acceptable, that is, $|\delta f| \leq 0.5$. Whenever the tracking error stays inside this band, the agent receives a constant positive reward $r = 1$. When the error magnitude exceeds the tolerance, i.e., $|\delta f| > 0.5$, but the force is still moving towards the new reference (indicated by $\text{sgn}(\delta f \cdot \delta f_d) \leq 0$,

where δf_d is the difference between the current and previous desired force), a moderate penalty proportional to the error and its rate is imposed: $r = -\alpha_1 \delta f - \alpha_2 \delta f'$, with α_1 and α_2 being positive weighting factors. Conversely, if the force deviates beyond the tolerance and overshoots the desired value (i.e., $|\delta f| > 0.5$ and $\text{sgn}(\delta f \cdot \delta f_d) > 0$), which reflects undesirable oscillatory behavior, we assign a significantly larger negative reward: $r = -\alpha_1 \delta f - \alpha_2 \delta f' - 100$.

For variable force tracking in RUSS, the overall procedure is jointly executed by the robot operator and the RUSS system as follows: (1) the operator communicates with the patient to localize the symptomatic region (no specialized US skills are required); (2) the maximum tolerable scanning forces for non-symptomatic and symptomatic areas (i.e., F_{Non} and F_{Sym} in Fig. 1) are determined by palpation, either manually or with robotic assistance; (3) during path planning, different desired force levels are assigned to different trajectory segments according to these limits; (4) the learned ORL-AIC controller is then used by the RUSS system to realize variable force tracking along the planned path. Note that the robot operator is typically not a professional sonographer, so the proposed RUSS framework can effectively relieve experienced physicians from repetitive scanning tasks. The complete training and deployment pipeline of ORL-AIC in the RUSS system is summarized in Fig. 2.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. Robot System Construction

An RUSS platform was constructed for real-world validation, as illustrated in Fig. 4. The system is built around a 7-DoF medical robot (Diana7-Med, Agile Robots, Germany), which is commanded from a C++ application developed under the Qt framework. A convex array probe (C5-1Xp, sSonic, China) is mounted at the robot flange through a customized mechanical adapter. A Nano Mini40 force/torque (F/T) sensor (ATI Industrial Automation, USA) is integrated between the probe and the adapter to provide real-time measurements of interaction forces and torques. US images are acquired from a commercial scanner (eHertzR Express, sSonic, China) via a VGA capture card (VGA41A-41+, JoinHope Image Inc., China) at a resolution of 1000×800 pixels and a frame rate of 20 Hz. All control and data acquisition modules run on a workstation equipped with an Intel i7-12700 CPU, 64 GB RAM, and an NVIDIA RTX 3080 GPU.

B. Real-world robotic policy learning and experimental validation

The dataset used for policy extraction can be gathered either online or offline. In principle, one may obtain samples by executing a random policy in the target domain. However, in robotic applications—particularly in clinical scenarios—online exploration with random actions can pose serious safety risks. Therefore, it is common to employ stabilizing controllers, obtained from demonstrations or a near-optimal policy, to safely generate demonstration data.

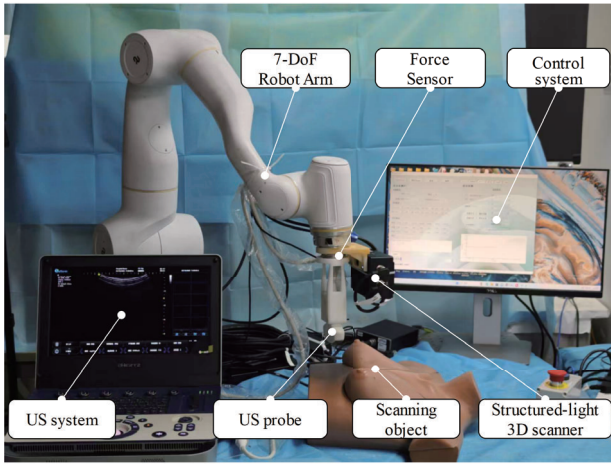


Fig. 4. Developed robot system for real-world validation.

In this study, we collected MDP-formatted samples on phantoms to guarantee safety, using stabilizing AIC controllers with different update gains η ($\eta \in [0.001, 0.005, 0.01, 0.05]$). Data were recorded from 40 distinct RUSS trajectories with randomly assigned target forces, where $F_{Non} \in [10, 20]$ and $F_{Sym} \in [3, 8]$. This procedure yields a dataset $D = \{(s_t, a_t, r_t, s'_t) \mid t = 1, 2, \dots, N\}$, which is used for learning the control policy and is assumed to consist of independent and identically distributed (i.i.d.) samples. During MDP sampling, when the RL agent selects an action a_t in state s_t , it receives a reward r_t and the dynamical system transitions to the next state s'_t .

Utilizing the data acquired above, we conducted training via the ORL algorithm, with the training procedure depicted in Fig. 3. Convergence was achieved after six iterations, satisfying the preset criterion ε of weight variation being less than 10^{-5} . The resultant weights were subsequently applied to our developed RUSS system as shown in Fig. 4. In order to rigorously assess the robustness of the proposed algorithm, we performed comparative evaluations of the variable force tracking performance between the ORL-AIC algorithm and four AIC algorithm under various parameter settings in both phantom and in vivo environments. The four comparative AIC methods are defined as AIC#1, AIC#2, AIC#3, AIC#4 corresponding to their η are set as 0.001, 0.005, 0.01, and 0.05, respectively. To meet the clinical requirements and the US scanning forces commonly used by physicians, we set $F_{Non} = 12$ N to ensure sufficient contact between the US probe and the skin in non-symptomatic area. In the symptomatic area, we set $F_{Sym} = 4$ N to satisfy the requirements of patient comfort and also US image quality.

1) *Phantom Study*: To validate the safety and tracking performance of the learned ORL-AIC policy, we conducted robotic variable force tracking experiments using a breast phantom made of silicone material, which features a complex contact surface. For comparison, we also tested the AIC algorithm with different hyperparameter along the same tracking trajectory. Table I demonstrates the force tracking performance

TABLE I
FORCE TRACKING RESULTS COMPARISON ON BREAST PHANTOM

Tracking policy	Setting time (s)	Overshoot (N)	Force error (N)
AIC#1 [6]	0.62 ± 0.38	0.85 ± 1.38	1.41 ± 0.34
AIC#2 [6]	1.13 ± 0.58	3.14 ± 3.37	1.68 ± 0.51
AIC#3 [6]	0.84 ± 0.37	1.99 ± 1.43	1.48 ± 0.35
AIC#4 [6]	1.03 ± 0.58	4.54 ± 4.06	1.87 ± 0.61
ORL-AIC	0.40 ± 0.21	0.69 ± 0.81	1.39 ± 0.32

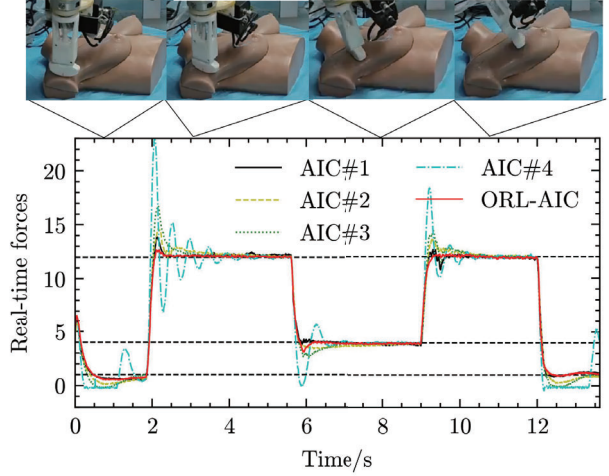


Fig. 5. Validation of ORI-AIC controller on breast phantom.

in terms of setting time, overshoot, and mean force error with three different trajectories. The results indicate that ORL-AIC significantly outperforms the AIC methods. Fig. 5 shows the variable force tracking curves of different methods under the same scanning trajectory, highlighting that the proposed ORL-AIC achieves lower overshoot and more stable tracking accuracy.

2) *In vivo Study*: Upper limb deep vein US scanning is an US-based procedure designed to evaluate the patency of the upper limb deep veins and detect the presence of thrombi, stenosis, or other pathological conditions. Motivated by the superior performance of ORL-AIC in phantom studies, we have extended its application to in vivo upper limb deep vein US scanning. Furthermore, we conducted a comparative analysis of four AIC methods (i.e., AIC#1, AIC#2, AIC#3 and AIC#4) under this procedure. The statistical results from three different scanning trajectories are presented in Table II. As indicated in Table II, the proposed ORL-AIC method exhibits favorable clinical applicability, characterized by minimal overshoot, which ensures the safety of the US scanning process, and high force tracking accuracy, as evidenced by the setting time and mean force error metrics. Notably, the hyperparameters of AIC are challenging to set for different patients due to variations in skin stiffness and shape. In contrast, ORL-AIC demonstrates superior tracking capability and adaptability without the need for tedious manual hyperparameter tuning. Fig. 6 depicts the force tracking trajectories of different force tracking methods in the upper limb deep vein US scanning

TABLE II
FORCE TRACKING RESULTS COMPARISON IN IN VIVO SCENARIO

Tracking policy	Setting time (s)	Overshoot (N)	Force error (N)
AIC#1 [6]	0.74 ± 0.37	1.71 ± 1.72	1.35 ± 0.43
AIC#2 [6]	1.03 ± 0.71	3.04 ± 3.08	1.61 ± 0.58
AIC#3 [6]	0.94 ± 0.56	2.19 ± 1.54	1.40 ± 0.40
AIC#4 [6]	1.08 ± 0.84	4.14 ± 3.42	1.63 ± 0.55
ORL-AIC	0.38 ± 0.16	1.19 ± 1.64	1.31 ± 0.40

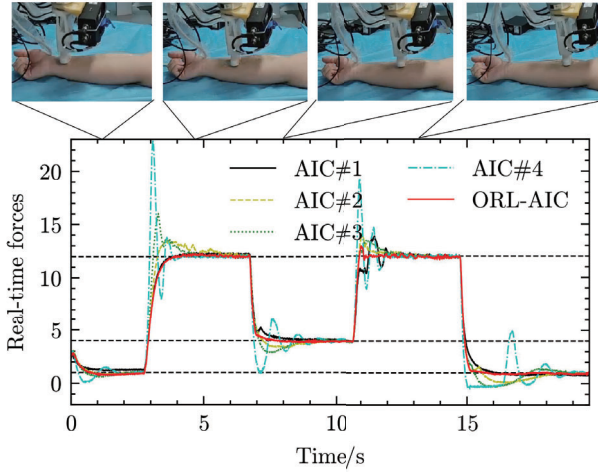


Fig. 6. In vivo validation of ORL-AIC controller in US arm scanning.

procedure. From Fig. 6, we can found that the proposed ORL-AIC method achieves high tracking accuracy to the target force with minimal deviation and also smaller overshoots, thereby fulfilling clinical requirements and ensuring patient comfort in a superior manner.

Remark: The phantom and in vivo experiments demonstrate that our ORL-AIC method can learn a high-precision force tracking policy using only 40 demonstration data samples. This highlights its applicability in clinical settings. Unlike online RL [10], our method does not require online interaction to learn policies, thereby effectively avoiding dangerous scenarios arising from unstable policy learning. In fact, during the initial learning phase, online RL primarily focuses on exploration, with its policy often leaning towards randomness. This makes the process of learning clinical policies of online RL highly risky. In contrast, our method exhibits stronger adaptability to medical scenarios which can learn a near-optimal policy from limited pre-collected samples.

V. CONCLUSION

Driven by the clinical requirements of robotic US scanning, a variable force tracking method based on ORL, namely ORL-AIC, is proposed in this paper, which learns from a limited number of pre-collected real-world robotic samples, thereby eliminating the need for real-time interaction inherent in online RL strategies. This characteristic is particularly advantageous for the application scenarios of force control in medical robots. Compared with traditional AIC methods, ORL-AIC

not only avoids cumbersome hyperparameter tuning but also better accommodates the unstructured environment caused by differences in skin stiffness and morphology among patients. Through phantom and in vivo experiments, the proposed ORL-AIC method exhibits better force tracking performance and clinical safety compared with traditional force tracking methods, which considers the individualized needs of patients in clinical applications and thus significantly enhancing the generalization performance of the RUSS system.

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